Problem

For the transistor network of Fig. 19.130,

$$h_{fe} = 80, \quad h_{ie} = 1.2\ \text{k}\Omega$$
$$h_{re} = 1.5 \times 10^{-4}, \quad h_{oe} = 20\ \mu\text{S}$$

Determine the following:

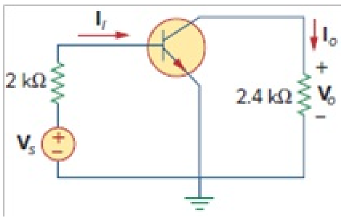
(a) voltage gain *Av* = *Vo⁄Vs*,

(b) current gain *Ai* = *Io⁄Ii*,

(c) input impedance Zin,

(d) output impedance Zout.

Figure 19.130



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

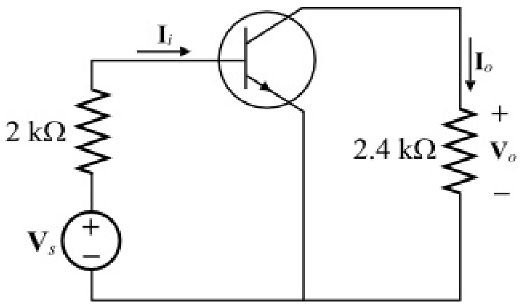


Figure 1

Step 2 of 5

(a)

Calculate the voltage gain of the transistor.

$$
\begin{aligned}
A_v &= \frac{V_o}{V_s} \\
&= \frac{-h_{fe}R_L}{h_{ie} + \left(h_{ie}h_{oe} - h_{re}h_{fe}\right)R_L} \\
&= \frac{-(80)\left(2.4\times10^3\right)}{1200+\left[(1200)\left(20\times10^{-6}\right)-\left(1.5\times10^{-4}\right)(80)\right]\left(2.4\times10^3\right)} \\
&= \frac{-192\times10^3}{1200+\left[0.024-0.012\right]\left(2.4\times10^3\right)} \\
&= \frac{-192\times10^3}{1200+\left[0.012\right]\left(2.4\times10^3\right)} \\
&= \frac{-192\times10^3}{1200+28.8} \\
&= \frac{-192\times10^3}{1228.8} \\
&= -156.25
\end{aligned}
$$

Step 3 of 5

Therefore, the voltage gain for the transistor, A_v is $\boxed{-156.25}$.

Step 4 of 5

(b)

Calculate the current gain of the transistor.

$$
\begin{aligned}
A_i &= \frac{I_o}{I_i} \\
&= \frac{h_{fe}}{1+h_{oe}R_L} \\
&= \frac{80}{1+\left(20\times10^{-6}\right)\left(2.4\times10^3\right)} \\
&= \frac{80}{1+0.048} \\
&= \frac{80}{1.048} \\
&= 76.33
\end{aligned}
$$

Therefore, the current gain of the transistor, A_i is $\boxed{76.33}$.

Step 5 of 5

(c)

Calculate the input impedance.

$$Z_{in} = h_{ie} - \frac{h_{re}h_{fe}R_L}{1+h_{oe}R_L}$$
$$= 1200 - \frac{(10^{-4})(80)(2.4\times10^3)}{1+(20\times10^{-6})(2.4\times10^3)}$$
$$= 1200 - \frac{19.2}{1+0.048}$$
$$= 1200 - \frac{19.2}{1.048}$$
$$= 1200 - 18.32$$
$$= 1181.68\ \Omega$$

Therefore, the input impedance, Z_{in} is $\boxed{1181.68\ \Omega}$.

(d)

Calculate the output impedance.

$$Z_{out} = \frac{R_s + h_{ie}}{(R_s + h_{ie})h_{oe} - h_{re}h_{fe}}$$
$$= \frac{2000+1200}{(2000+1200)(20\times10^{-6})-(1.5\times10^{-4})(80)}$$
$$= \frac{3200}{0.064-0.012}$$
$$= \frac{3200}{0.052}$$
$$= 61.54\ \text{k}\Omega$$

Therefore, the output impedance, Z_{out} is $\boxed{61.54\ \text{k}\Omega}$.